Aditya Manoj

Developer I – Full-Stack, AI Systems & DevOps Engineering

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Professional Profile

Dynamic Computer Science Engineer with **2+ years** of experience architecting production full-stack systems, multimodal AI desktop agents, zero-knowledge security architectures, and automated CI/CD infrastructure. Proven track record engineering cross-platform (macOS/Linux/Windows) real-time streaming pipelines, offline-first PWAs, and high-concurrency event operating systems processing **1.2M+ operations** with 100% uptime. **Meta-Certified Full Stack Developer** specializing in asynchronous Python systems, low-latency audio/GUI rendering (PyQt6), WebSocket protocols (Gemini Live), local LLM inference (Ollama), Docker, and edge cloud platforms (Firebase, Vite CDN). Dedicated to designing resilient, privacy-first, and **high-impact software architectures**.

DevOps & Technical Skills

- **DevOps & Cloud Infrastructure:** Python (Asyncio/OOP) — Git/GitHub Workflows — CI/CD Pipelines — Bash/PowerShell — Docker — Firebase (Hosting, Cloud Functions, Firestore) — Vite Edge CDN
- **Multimodal AI & Systems:** Gemini Live API (Full-Duplex WebSockets) — Local LLMs (Ollama/vLLM) — Model Context Protocol (MCP) — Agentic Specs / Prompt-Ops — RapidOCR / OpenCV — Cross-Platform APIs (Win32, macOS AppKit, Linux DBus)
- **Frontend & Desktop Engineering:** React / React Native — PyQt6 GUI (Software/OpenGL) — JavaScript (ES Modules) — IndexedDB (Dexie.js) — WebGL / Three.js / Canvas — PortAudio / SoundDevice — WebAuthn / Passkeys

Certifications & Specializations

Meta Full Stack Developer Specialization *2024–2025*
Coursera (Meta)

- Full-stack mastery covering React, Python/Django, REST APIs, database design, HTML5/CSS3 and **production deployment**.

Professional Experience

Full-Stack PWA & Security Systems Engineer *Jan 2026 – Present*
SBAFA Financial Enclave (sbafa-ft.web.app) *Remote*

- Architected an **offline-first**, **zero-knowledge** Progressive Web Application (PWA) utilizing vanilla ES modules, IndexedDB (Dexie.js), and Firebase Firestore, orchestrating bi-directional reactive synchronization for 1,000+ ledger records.
- Engineered an on-device, sandboxed PDF statement ingestion engine leveraging `pdfjs-dist` and deterministic cryptographic passkey decryption, executing multi-bank **OCR parsing** with zero third-party telemetry.
- Implemented an automated OAuth2 Gmail pipeline to ingest and decrypt scheduled banking e-statements, deploying contextual regex heuristics for **zero-latency** transaction deduplication.
- Hardened client security and session durability via **WebAuthn passkeys**, SHA-256 PIN hashing, and multi-tier Service Worker caching strategies (`Stale-While-Revalidate`) to sustain offline financial tracking.

Lead PWA & Frontend Systems Engineer *Jul 2026 – Sep 2026*
IndZita Biotech (indzitabiotech.com) *Remote*

- Architected an **offline-first** Progressive Web Application (PWA) leveraging custom Service Worker strategies and IndexedDB caching to serve a high-performance **3D visualization suite** (Catmull-Rom splines, Euler kinematics) with zero load-time overhead.
- Built an installable, mobile-optimized **PWA app shell** and reactive genomic canvas, optimizing frame-budget lifecycles via `IntersectionObserver` to sustain **fluid 60 FPS performance** across low-spec handheld devices.
- Engineered an offline-capable Virtual **Diagnostics Lab simulator** modeling real-time thermal PCR kinetics (C_t thresholds) and piRNA biomarker workflows, translating high-stakes requirements for clinical partners **(RGCB, Karkinos Healthcare, NIMHANS)** under the MAHA MedTech Mission.
- Provisioned PWA web app manifests, background sync handlers, and automated Vite edge deployment pipelines on Firebase Hosting, achieving **sub-second global delivery** and **zero-downtime reliability**.

Full-Stack & DevOps Engineer *Apr 2026 – Jun 2026*
Threadpath Collective (threadpathcollective.ca) *Remote*

- Architected end-to-end **full-stack web applications** using React and Node.js microservices, engineering **scalable API integrations** and state management across complex user workflows.
- Developed serverless backend logic using Firebase Cloud Functions, implementing robust **payload validation**, error-handling middleware, and optimized **database queries**.
- Configured automated **CI/CD build pipelines** for frontend compilation and backend serverless endpoints.

- Diagnosed and resolved domain-level CDN caching discrepancies across custom endpoints to guarantee **zero-downtime releases**.

Lead Developer (Technical Trainee) *June 2025 – Nov 2025*
Rajiv Gandhi Centre for Biotechnology – IAN-2025 (ian2025.in) *Thiruvananthapuram*

- Built and deployed a Progressive Web App (PWA) managing on-ground logistics for an international neuroscience conference, achieving **100% uptime** during peak volume.
- Automated event workflows: processed **1.2M reads** on Firestore and engineered a digital authentication system executing 2,275+ food coupon scans with zero errors.
- Maintained secure **access controls** (`IAN/2025/J21`) to isolate sensitive logistical workflows and conference parameters during live execution.
- Developed Python **automation scripts** and data integration routines following strict OOP principles throughout the codebase.

Product Engineering Intern *Jan 2025 – Aug 2025*
Lascade LLP – Travel Animator (travelanimator.com) *Kochi*

- Spearheaded UI design enhancements and full-stack performance optimizations for **Travel Animator** (travelanimator.com), delivering highly responsive, interactive UI components and intuitive user flows.
- Engineered asset loading pipelines and lightweight animation workflows, significantly reducing **page latency** and optimizing rendering efficiency across low-bandwidth environments.
- Managed Git branching workflows, peer pull-request reviews, and collaborated on **CI/CD pipelines** for automated application builds and staging deployments.

AI/ML Intern *Jul 2024 – Oct 2024*
ICT Academy of Kerala *Thiruvananthapuram*

- Engineered Python data preprocessing pipelines that boosted image classification accuracy by **10% gain** using advanced data augmentation techniques.
- Fine-tuned facial recognition models via supervised learning, achieving a **5% reduction** in false positive detections.

App Developer & Technical Mentor *Oct 2023 – Mar 2024*
Sarvodayam VHSS Arayampadam *Thrissur*

- Co-developed the **Ente Naddu** mobile application backed by the Government of Kerala for school students over a 6-month development cycle.
- Mentored student developers across requirement gathering, system architecture, feature implementation, and **code optimization**.

Full Stack Developer Intern *May 2023 – Jul 2023*
ICT Academy of Kerala *Thiruvananthapuram*

- Architected a responsive MERN stack application, executing full deployment transition from local environments to **cloud infrastructure**.
- Integrated Firebase real-time database listener structures for multi-user data synchronization and **high availability**.

Technical Projects

ALFRED Mark-VII – Autonomous Multimodal Desktop Assistant *github.com/AdityaManojA/ALFRED-MK-V*

- Architected a **cross-platform (macOS/Linux/Windows)** multimodal desktop agent with sub-second full-duplex voice/vision streaming using Google Gemini Live WebSockets, fallback local LLM engines (Ollama/vLLM), and **35+ autonomous tool integrations**.
- Engineered a tactical **PyQt6 HUD interface** with hardware-independent software rendering, sub-50ms audio jitter buffering via PortAudio, and optimized frame pacing, slashing render latency by **82% (125ms → 22ms)**.
- Implemented hybrid visual grounding (RapidOCR, spatial click coordinate mapping), background Sentry distraction defense, and local zero-telemetry execution pipelines to enforce strict structural data privacy.

Agentic Prompt-Ops & System Specs Engine *github.com/AdityaManojA/prompt-ops-blueprints*

- Engineered structured **agentic blueprints** and technical specifications to guide autonomous coding agents (Cursor, Windsurf, Claude Code) in generating production-grade architectures.
- Boosted **framework efficiency** across 8+ tech stacks (Next.js, Supabase, Cloudflare Workers, SvelteKit) by enabling LLMs to parse system constraints and interface with Model Context Protocols (MCPs).
- Standardized developer **agentic workflows** by replacing ambiguous natural language instructions with deterministic implementation plans and multi-agent execution graphs.

Education

B.Tech, Computer Science & Engineering *2021 – 2025*
Jyothi Engineering College *Thrissur*

Achievements & Leadership

- Appointed by Rajiv Gandhi Centre for Biotechnology to lead core digital infrastructure, supporting 2,200+ secure interactions with **100% uptime**.
- Awarded official **Certificate of Appreciation** by Jyothi Engineering College / Govt. of Kerala for technical leadership and mentorship.
- Recognized for **technical excellence** and lab contribution under Dr. Jackson James at the Rajiv Gandhi Centre for Biotechnology.